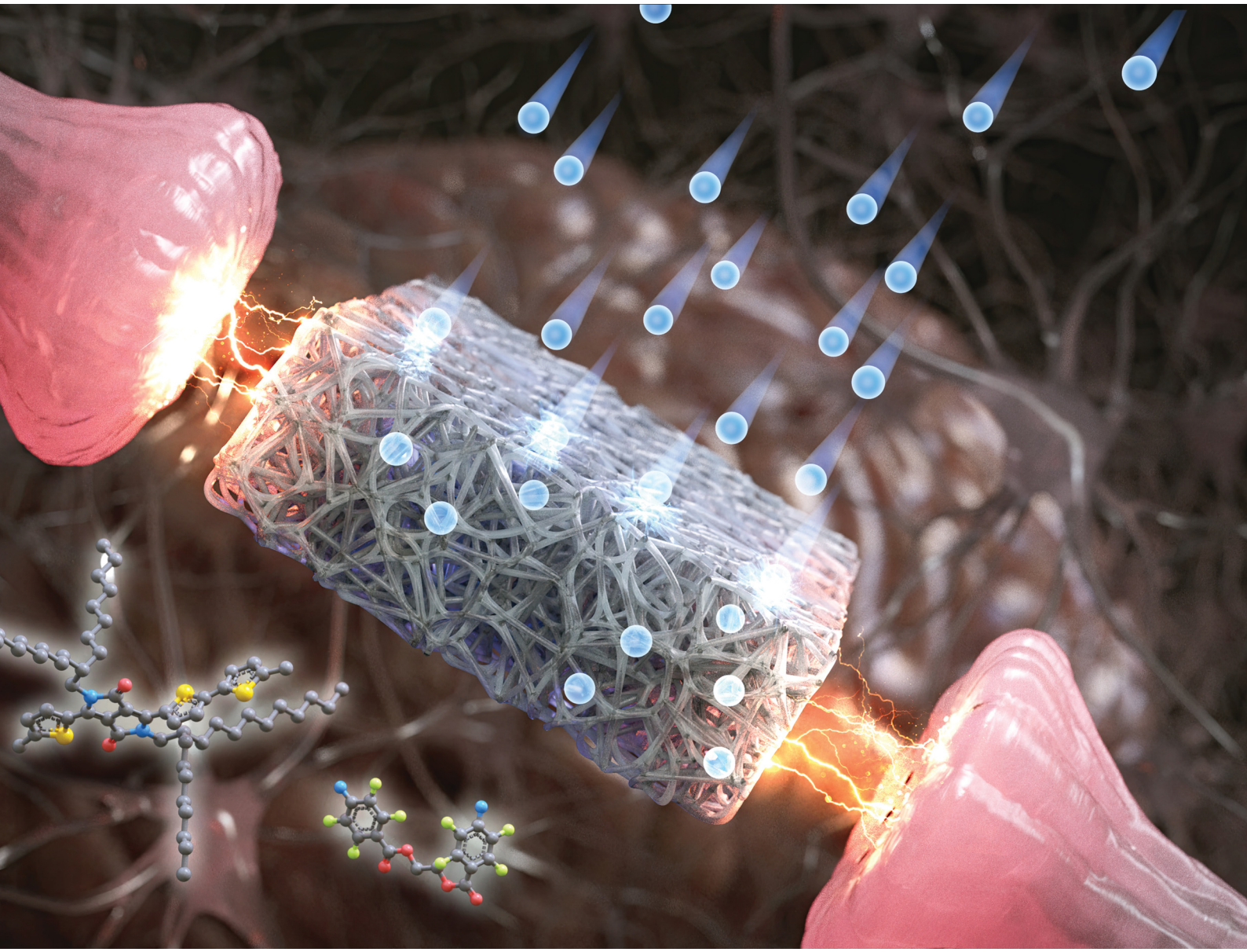
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Crosslinking-induced anion transport control for enhancing linearity in organic synaptic devices†

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Numerous studies on neuromorphic computing systems and their associated synaptic devices have been reported for the efficient processing of complex data. Among them, organic electrochemical transistors (OECTs) have attracted considerable attention owing to their advantages such as low cost, high scalability, and facile electrical modulation. However, the requirement of supplementary processing for ionic transport control to actualize or enhance synaptic attributes necessitates a compromise between their inherent benefits. Here, we developed a simple method, photoinduced crosslinking, which can control the structure of conjugated polymers in OECTs to improve ionic transport control. Crosslinked polymers increase the ion doping efficiency and allow sequential anion movements, which leads to high linearity in OECTs. The fabricated device also exhibited enhanced synaptic properties such as a long retention time, wide dynamic range, and high recognition accuracy. This innovative approach opens up new possibilities for the construction of next-generation artificial synapses.

New concepts

OECTs can realize neurotransmitter release through the transport of ions, so it is important to control the iontronics in OECTs to simulate various synaptic functions. To achieve this goal, conventional OECTs require ionic control layer design or additive processing, which increases cost and limits solvent selection. In this work, we modulated the structure of conjugated polymers through crosslinking density control *via* click chemistry-designed photo-induced crosslinking and effectively controlled the ion transport properties of organic synaptic devices. With proper ion transport control, the artificial synapses achieved wide nonvolatile current control and high linearity and recorded high accuracy in neural network simulations. The novel approach of structural modulation *via* photocrosslinking for controlling ion transport properties lowers the threshold for fabricating high-performance organic synaptic devices and offers sufficient promise for realizing next generation neuromorphic computing.

Introduction

Organic semiconductor transistors have attracted attention owing to their advantages, such as low cost,[1,2] high scalability,[3–5] and facile electrical modulation through the easy tunability of organic molecules.[6,7] In particular, organic electrochemical transistors (OECTs) which regulate the electrochemical doping state of organic channels through bias applied to the electrolyte, have attracted attention because they exhibit excellent properties such as low operating voltage and large transconductance.[8–10] These characteristics originate from the field effect in OECTs that occurs over the entire area of the channel as ions accumulate, whereas the field effect in thin-film transistors occurs in a narrow region at the dielectric–semiconductor interface.[11,12] Because of the volumetric doping of the electrolyte ions, OECTs are capable of larger drain current modulation at lower voltages. Furthermore, biological ion transport-based neural mechanisms can be implemented through changes in channel conductance due to the movement of electrolyte ions. The biological brain performs learning and memory retention through changes in synaptic weights due to neurotransmitter movement, which can provide guidance for implementing various biological functions in artificial synapses based on synaptic plasticity, as well as serve as the basis for biomimetic systems with perceptual capabilities for different forms of stimuli.[13–15] In this respect, OECTs hold promising possibilities for neuromorphic computing.[16]

In OECTs, properly controlled ionic transport properties prevent rapid accumulation or dissipation of ions under an electrical field, leading to efficient volumetric doping and linear conductance updates.[17] This improves the accuracy of learning and recognition in synaptic devices, which is beneficial for their applications in neuromorphic computing. Therefore,

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efforts have been made to improve the ionic transport properties of organic channels to fabricate highly reliable devices beyond the implementation of synaptic properties. Among these, the development of conjugated polymer-based synaptic devices has attracted considerable attention. Conjugated polymers have emerged as promising channel materials due to the advantages of less conformational changes for the movement of ions and the ability to control ion transport properties based on their chemical structure,[18] such as poly(3-4-ethylenedioxythiophene): poly (styrene-sulfonate) (PEDOT:PSS), poly(3,4-ethylenedioxy thiophene):tosylate (PEDOT:Tos)/polytetrahydro furan (PTHF), poly(3-hexylthiophene-2,5-diyl) (P3HT), and others are being investigated for devices based on these materials.[19–24] There are various attempts to improve ionic doping efficiency and realize high-performance synaptic transistors such as adding dopants for ion doping,[22,25,26] film structure modification,[23,27,28] and additional interlayer design.[19] However, these parameters increase the complexity of the process and degrade the intrinsic properties of the polymer, which are potential limitations to its application in next-generation neuromorphic computing. Therefore, a simpler method to enhance the ion doping efficiency is required.

The excellent tunability of conjugated polymers offers the possibility of performance enhancement through molecular modeling using moiety and side chain engineering. In OECTs, properties such as transconductance, morphology, and operating stability can be controlled by side chain modulation, and numerous side chain configurations can be considered.[29] Among these, hydrophilic side chain groups, including ethylene glycol, are considered promising candidates because they can increase the ionic conductivity of the thin films, leading to efficient ion doping.[30–33] Various groups have shown that OECTs can exhibit good performance and stability by modulating the hydrophilic side chains[34–38] or controlling their ratio with alkyl chains.[39–41] In particular, introducing hydrophilic side chains into conjugated polymers with alkyl chains can be an effective way to fabricate OECT-based synaptic devices with high ion-doping efficiency, because alkyl chains act as spacers that provide space for ions to penetrate without hindering charge transfer.[41] Several groups have proposed synaptic devices with hydrophilic side chains introduced into conjugated polymers containing alkyl side chains.[42–44] However, these methods demand time and effort because they require polymer synthesis steps to attach or replace hydrophilic side chains. In addition, the low solubility of the hydrophilic side chains in organic solvents[34,35,42,45] results in a trade-off in crystallinity[46] and provides a narrow solvent selection.[37] Therefore, solvent-insensitive structural modulation of conjugated polymers should be proposed to select only the benefits of introducing hydrophilic side chains.

Herein, we propose conjugated polymer-based synaptic transistors with improved ion transport and nonvolatile properties by introducing hydrophilic side chains through click chemistry-designed photocrosslinking. Previous studies have shown that crosslinking agents can alter the microstructure of organic channels to improve their durability and modulate their ion transport properties.[47] A nitrene-induced photocrosslinker (NIPS) formed crosslinked structures between the alkyl spacers of poly(diketopyrrolopyrrole–terthiophene) (PDPP3T) upon UV exposure.[48] The oligo(ethylene glycol) introduced by NIPS enhanced ion doping and effectively controlled the accumulation and dissipation of anions in the channel layer. With sufficient ion feeding control, our PDPP3T/NIPS-based synaptic devices exhibited 52% lower energy consumption ($\sim$0.59 pJ) compared to PDPP3T-based devices based on a lower operating voltage and maintained a high nonvolatile current for 10^2 s. In addition, they exhibited low device variation and stable linear weight updates over 5000 electrical stimulations while implementing long-term potentiation/depression (LTP/D) properties. Based on the remarkable property enhancement of the PDPP3T/NIPS-based synaptic devices, high accuracies of 89% and 90% were achieved in object group classification and handwritten recognition simulations, respectively. These results provide a novel method for conjugated polymer structure modulation to improve synaptic properties and demonstrate its potential for the design of enhanced artificial synapses.

Results and discussion

Fig. 1a shows the structures of a biological neuronal synapse and an artificial synapse based on OECTs. Presynaptic neurons generate action potentials in the form of spikes that release neurotransmitters. The released neurotransmitters reach the end of the presynapse and are transmitted through receptors at the postsynapse. When neurotransmitters are transmitted to the postsynapse, postsynaptic currents (PSCs) are generated during ion exchange, the magnitude of which depends on the number of transmitted neurotransmitters. Synapses perform various functions, including memory strengthening and forgetting, by modulating the magnitude of PSCs in response to spikes to appropriately change the synaptic weights. Our OECT-based artificial synapses, which were fabricated to mimic biological synapses, consisted of a gate electrode through which a presynaptic pulse was applied, a source and drain electrode acting as a postsynapse, a PDPP3T/NIPS channel polymer acting as a receptor, and a diethylmethyl(2-methoxyethyl)ammonium bis(trifluoromethyl-sulfonyl)imide (DEME-TFSI) ionic liquid to implement neurotransmitters. The device was fabricated by spin-coating the PDPP3T/NIPS blend onto a SiO_2/Si substrate, UV crosslinking for 5 s, and deposition of Au electrodes using a thermal evaporator (Fig. S1, ESI†). We used PDPP3T-octa-dodecyl (OD), which is composed of two OD side chains attached to a PDPP3T backbone that can easily control the structure and has been shown to change synaptic properties depending on the side chain length.[49] We crosslinked it with NIPS consisting of oligo(ethylene glycol) with perfluorophenyl azide coupled at both ends to control ionic transport (Fig. 1b). We selected 3-NIPS with three oligo(ethylene glycol) repeats. The molecular structure of DEME-TSFI used as the electrolyte is shown in Fig. S2, ESI.† As shown in Fig. 1c, the bare PDPP3T and PDPP3T/NIPS blends were completely soaked and erased in chlorobenzene, whereas the NIPS-crosslinked PDPP3T did not soak and maintained the shape of the film. To further observe the changes caused by

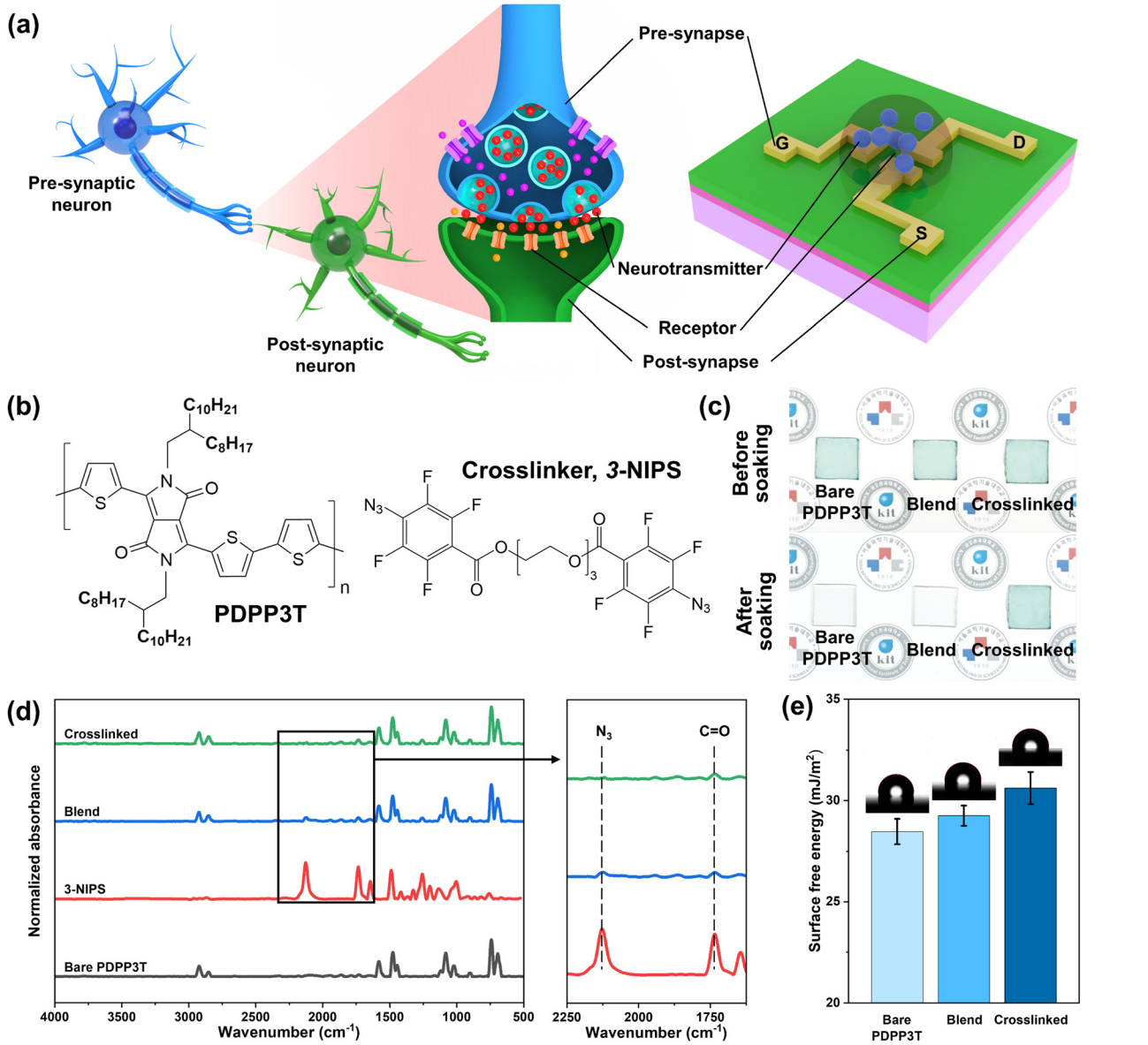


Fig. 1 (a) Structure of a biological neuron synapse and OECT based artificial synapse. (b) Structure of PDPP3T and a 3-NIPS crosslinker. (c) Images of spin-coated PDPP3T, PDPP3T/NIPS blend, and crosslinked PDPP3T/NIPS on a quartz substrate before and after soaking in chlorobenzene. (d) FT-IR spectra of PDPP3T, 3-NIPS, blended PDPP3T/NIPS, and crosslinked PDPP3T/NIPS. (e) Surface free energy of PDPP3T, PDPP3T/NIPS blend, and crosslinked PDPP3T/NIPS.

NIPS crosslinking, we measured the energy absorbed by the vibrations and rotations of the molecular structure using Fourier-transform infrared (FT-IR) spectroscopy (Fig. 1d). Two peaks (2170, and 1740 cm^{-1}) were clearly observed in the PDPP3T and PDPP3T/NIPS blends, owing to the N_3 resonance and C═O double bond contained in NIPS. However, the N_3 resonance peak was removed after crosslinking, whereas the C═O double bond peak was retained. This shows that a change occurs in the N_3 resonance bond of NIPS after UV crosslinking; that is, before UV treatment, PDPP3T and NIPS exist as separate substances in the mixture. However, after UV treatment, the photocrosslinker termini change and form bonds with PDPP3T (Fig. S3, ESI†). The solvent soaking test and FT-IR spectroscopy results showed that the crosslinking of PDPP3T and NIPS was successful, resulting in changes in their properties.

To confirm that the surface energy of the film changed with NIPS crosslinking, we performed contact angle measurements (Fig. 1e). To calculate the surface energies of the films in the polar and nonpolar regions, we used two solvents: deionized water (DIW) and diiodomethane (DIM). When DIW was used, the contact angles on the film were 104° for PDPP3T, 99.4° for the PDPP3T/NIPS blend, and 96.7° after crosslinking, indicating stronger attraction between the solid surface and liquid. We also found that the contact angle was significantly reduced even in the nonpolar region when using DIM: 58.8° for PDPP3T, 60.1° for the blend, and 56.9° after crosslinking. We calculated the surface

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free energy (γ_s) of the film using the equation given below.[50]

$$\frac{\gamma_l(\cos\theta+1)}{2}=\sqrt{\gamma_l^n}\sqrt{\gamma_s^n}+\sqrt{\gamma_l^p}\sqrt{\gamma_s^p}$$

γ_l^n and γ_l^p are the nonpolar and polar components in the liquid region, respectively, and γ_s^n and γ_s^p are the nonpolar and polar components in the solid region (the values for each component are described in Table S1, ESI†), respectively. The calculation results show that the surface free energies of PDPP3T, the PDPP3T/NIPS blend, and crosslinked PDPP3T/NIPS are 28.46 ± 0.63 mJ m^{-2}, 29.25 ± 0.50 mJ m^{-2}, and 30.62 ± 0.80 mJ m^{-2}, respectively, confirming that the surface free energy of the channel is enhanced by the addition of NIPS. In OECTs, the contact angle between the electrolyte and channel polymer indicates interfacial compatibility, and an increase in compatibility with a decrease in the contact angle owing to an increase in surface free energy is favorable for polarizing and depolarizing carriers.[51] This implies that the hydrophilic crosslinking agent increases the ion doping efficiency of the device. Furthermore, we performed 2D grazing-incidence wide-angle X-ray scattering (GIWAXS) analysis to observe how the crystallinity of the films changed after UV crosslinking, and the GIWAXS mapping results showed that mixing or crosslinking NIPS did not significantly deteriorate the crystallinity (Fig. S4, ESI†).

To observe the effects of improving the ion transport properties *via* NIPS crosslinking on the electrical characteristics of the devices, we first examined the transfer curves (Fig. S5, ESI†). The devices with and without NIPS exhibited clockwise hysteresis windows with a gate voltage applied from 2 V to −3 V, which are characteristics when mobile ions in ionic liquids are more dominant in window formation. After the crosslinking of NIPS, the threshold voltage decreased from −2 V to −1.5 V, indicating that ion-doping was enhanced by the combination of NIPS, allowing the device to operate with weaker stimuli. To determine how crosslinked NIPS causes changes in synaptic behavior, we measured the plasticity properties of PDPP3T and PDPP3T/NIPS-based devices. The short-term plasticity (STP) of PDPP3T and PDPP3T/NIPS-based devices upon application of single pulses of varying amplitudes at a drain voltage (V_{DS}) of −1 V is shown in Fig. S6, ESI.† Both devices showed an excitatory postsynaptic current (EPSC) proportional to the presynaptic voltage (V_{Pre}) amplitude, followed by a gradual decrease in the channel current for a short period. The EPSC of both devices reached maximum values of 1.02 μA and 0.29 μA at −2 V, respectively, which is like the effect of strengthening synaptic connections in the human brain as the intensity of input spikes increases. The 38% longer transient time in the NIPS crosslinked device is noteworthy because it suggests that NIPS inhibits the departure of mobile ions, allowing stimuli to be remembered for longer.

To characterize long-term plasticity (LTP), 10 consecutive pulses (V_{Pre} = −1 to −2 V, −0.25 V interval) with a width and interval of 80 ms were applied at a constant V_{DS} of −1 V (Fig. 2a and b). Both devices showed a continuous increase in the EPSC values with consecutive pulse inputs. Immediately after the

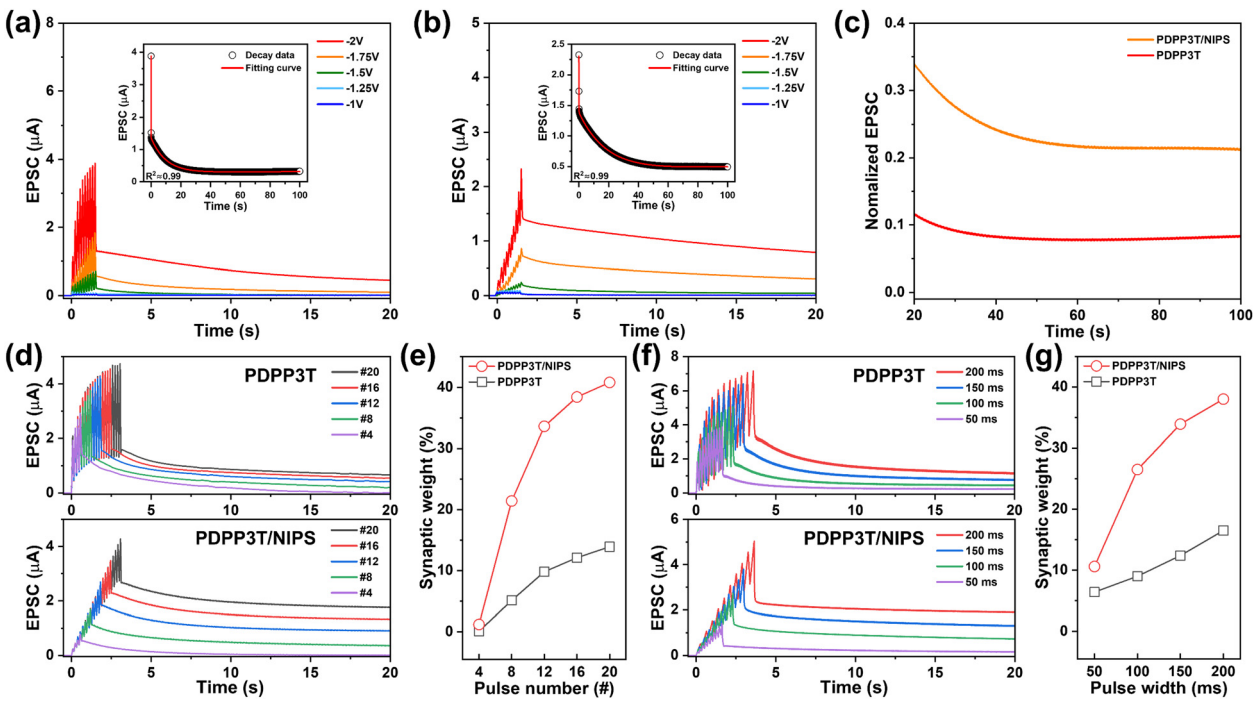


Fig. 2 Long-term plasticity characteristics of (a) PDPP3T and (b) PDPP3T/NIPS by applying 10 consecutive pulses of different amplitudes (V_{Pre} = −1 to −2 V, −0.25 V interval) with a width and delay of 80 ms at V_{DS} = −1 V. Inset: tri-exponential fitting curves of decay. (c) Normalized EPSC of PDPP3T and PDPP3T/NIPS based devices at V_{Pre} = −2 V. (d) EPSC of PDPP3T (upward) and PDPP3T/NIPS (downward) with different numbers of consecutive pulses applied at V_{DS} = −1 V, V_{Pre} = −2 V, and (e) Retained synapse weight at 20 s. (f) EPSC of PDPP3T (upward) and PDPP3T/NIPS (downward) with different widths of consecutive pulses applied at V_{DS} = −1 V and V_{Pre} = −2 V, and (g) retained synaptic weight at 20 s.

pulse was removed, PDPP3T showed a decrease of 69% of its maximum EPSC (3.88 μA), while PDPP3T/NIPS showed a decrease of only 38% of its maximum (2.32 μA). Subsequently, the EPSC of both devices showed a gradual decrease, resulting in a residual current of 12% for PDPP3T and 34% for PDPP3T/NIPS after 20 s. To determine the proportion of nonvolatile current, we checked the retention behavior for up to 10^2 s (Fig. 2c). PDPP3T retained ~8% nonvolatile channel current after 40 s, with an unstable behavior of increasing current after 70 s. PDPP3T/NIPS, on the other hand, showed excellent characteristics, retaining ~21% nonvolatile channel current after 60 s and up to 10^2 s. To analyze the trend of LTP measured in both devices, we applied a tri-exponential model that separated the PSC decay region into three parts. Immediately after the application of 10 consecutive pulses, the decrease in current from the EPSC peak was calculated as

$$I = I_1 \exp\left(-\frac{t}{\tau_1}\right) + I_2 \exp\left(-\frac{t}{\tau_2}\right) + I_3 \exp\left(-\frac{t}{\tau_3}\right) + I_0$$

where I_0 is the base current before the pulse is applied and I_1, I_2, and I_3 are the pre-factors of the PSC. τ_1, τ_2, and τ_3 are the time constants in each decay region, representing depolarization of EDL, ion depletion in the amorphous region, and ion depletion in the crystalline region, respectively.[28] PDPP3T/NIPS recorded higher τ values compared to PDPP3T in all regions, and the ion depolarization time in the crystalline region was clearly increased from 58 s to 161 s. This indicates that the crosslinked NIPS effectively inhibited the back-diffusion of TFSI anions, such that the adsorbed ions remained in the channel for a longer time. Through the control of ionic transport by NIPS, the device exhibited enhanced nonvolatile characteristics with longer transient times. Owing to these synaptic properties' enhancement, our devices offer the possibility of reversible write and erase processes and multibit operations for neuromorphic computing.

To further confirm the effect of the crosslinked NIPS on the EPSC and nonvolatile current properties, we observed the nonvolatile current enhancement characteristics as a function of the pulse conditions. First, we measured the EPSC variation (Fig. 2d) and the synapse weight residual at 20 s (Fig. 2e) as a function of the number of pulses with an applied width and interval of 80 ms and an amplitude of −2 V at V_{DS} = −1 V. We found that the EPSC varied with the number of pulses with an applied width and interval of 80 ms. The EPSC and residual synaptic weight increased as the number of consecutive pulses applied to both devices increased. This is similar to the switch from short-term memory (STM) to long-term memory (LTM) in the human brain as the strength and frequency of repeatedly applied stimuli increase. However, PDPP3T showed a 13.8% increase in retention at 20 s, from 0.1% (four pulses) to 13.9% (20 pulses), whereas PDPP3T/NIPS showed a 39.6% increase, from 1.2% (four pulses) to 40.8% (20 pulses), indicating a more effective transition from STM to LTM. Fig. 2f and g shows the EPSC and the synapse weight residual at 20 s when 10 pulses of different widths with an interval of 80 ms were applied under V_{DS} = −1 V. Similar to the results in Fig. 2d and e, the synapse weight as a function of pulse width was more retained in PDPP3T/NIPS compared to PDPP3T, increasing by 27.4% and 10.1%, respectively. The above results demonstrate that controlling the ionic transport *via* NIPS crosslinking can enhance the nonvolatile properties of the device, enabling a stronger memory of the applied stimulus for a longer period.

To analyze why the nonvolatile properties increased with the crosslinking of NIPS, we performed X-ray photoelectron spectroscopy (XPS) to determine the binding energy changes upon pulse application. Fig. 3a shows the C1s spectra before and after ten consecutive pulses were applied to the PDPP3T channel. The high intensity of the C═C/C–C bond was due to the alkyl chains comprising the side chains and did not change significantly after the pulses were applied. The CF_3 bond at 293 eV originates from the TFSI anion, proving that part of the anion accumulated at the electrolyte/channel interface is adsorbed on the channel polymer upon pulse application. This trend is similar to that observed for the PDPP3T/NIPS channel (Fig. 3b). The area occupied by C–O bonds in the PDPP3T/NIPS channel increased under the influence of the oligo(ethylene glycol) contained in NIPS. One point of interest is the percentage of the area occupied by the CF_3 bonds in both devices. After 10 consecutive pulses, the area occupied by CF_3 in the PDPP3T channel was only 2.31%, whereas it reached 5.70% in the PDPP3T/NIPS channel (Fig. 3c and d). This indicates that the crosslinking of NIPS can induce more TFSI anions. For further confirmation, we analyzed the O1s spectra to determine the extent of TFSI anion adsorption after applying the pulses. The SO_2 bond located at 533 eV is unique to TFSI anions, as is the CF_3 bond, making it a good candidate for determining whether and how strongly anions are adsorbed. The PDPP3T channel exhibited one simple peak because no oxygen bonds, except O═C, were present (Fig. 3e). Therefore, the O–C/C–O–C and O═C–O bonds shown in Fig. 3f originate from NIPS. Fig. 3g-h shows the area ratios of the bonds in the O1s spectra. The TFSI anions that accumulated upon the negative pulse application appeared as SO_2 peaks. The area ratio occupied by other bonds decreased with an increase in the SO_2 peak area ratio, and similar to the results in C1s, the area ratio occupied by SO_2 bonds in the PDPP3T/NIPS channel was 16.59%, which is higher than the 9.64% calculated in the PDPP3T channel. The results in Fig. 3a–h show that the introduced NIPS is advantageous for the effective adsorption of TFSI anions and supports previous electrical data results.

Fig. 3i and j illustrate the LTP mechanism of the PDPP3T and PDPP3T/NIPS-based devices. In the initial equilibrium state, PDPP3T/NIPS exhibited a lower contact angle with the applied gate voltage. DEME cations migrated to the electrolyte/gate interface, and TFSI anions migrated to the electrolyte/channel interface and accumulated. Some of the accumulated anions migrate across the interface into the channel layer and can be adsorbed onto appropriate sites. The PDPP3T/NIPS device allowed more ions to enter the channel layer because of its high interfacial compatibility. At the end of the pulse sequence, rapid depolarization occurs, allowing anions to accumulate at the electrolyte/channel interface and the channel layer to diffuse back into the electrolyte and return to the initial

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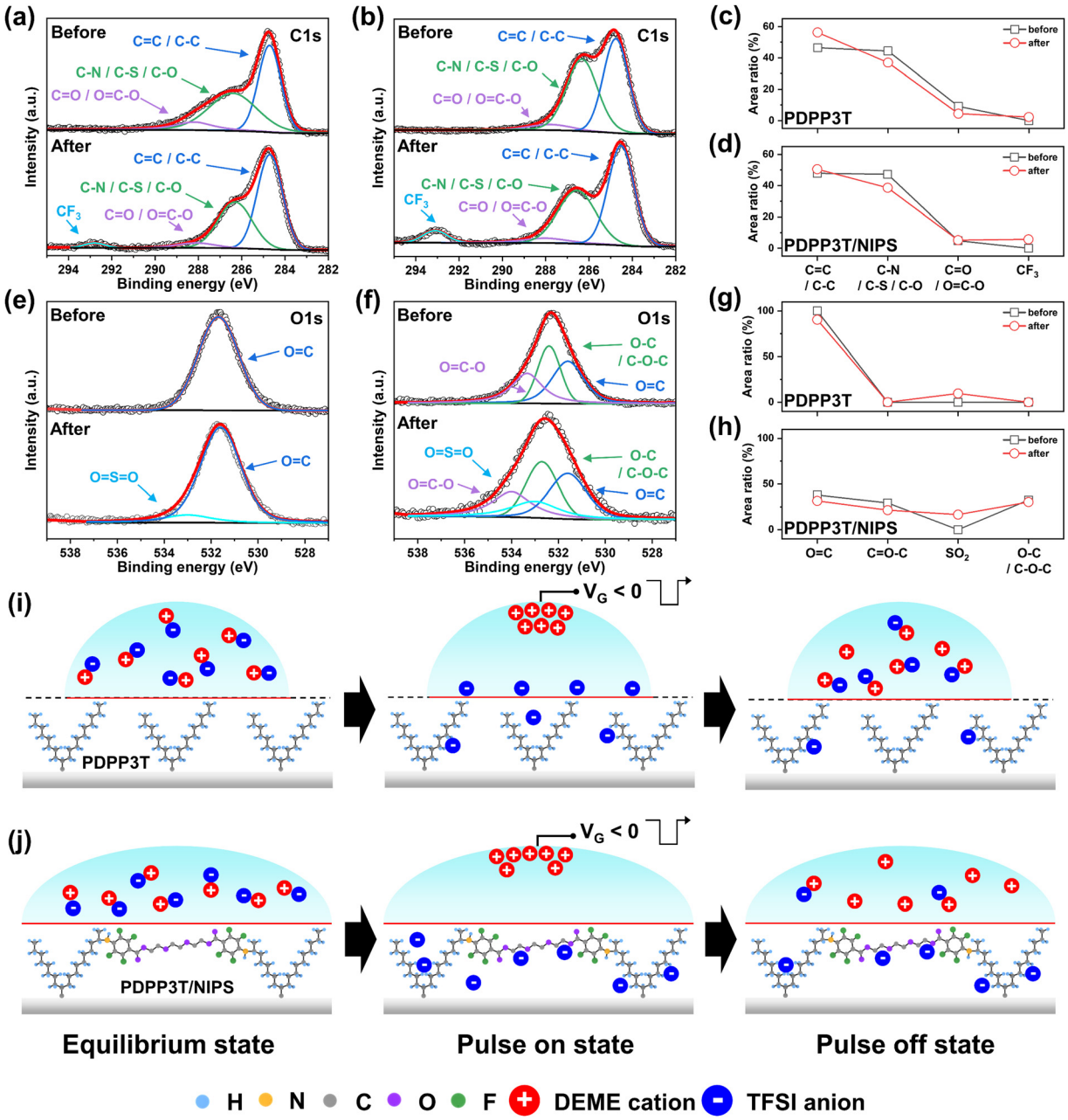


Fig. 3 (a) C1s XPS spectra of PDPP3T based devices before and after applying 10 consecutive pulses. (b) C1s XPS spectra of PDPP3T/NIPS based devices before and after applying 10 consecutive pulses. C1s XPS peak area ratios in (c) PDPP3T and (d) PDPP3T/NIPS based devices with and without consecutive pulse application. (e) O1s XPS spectra of PDPP3T based devices before and after applying 10 consecutive pulses. (f) O1s XPS spectra of PDPP3T/NIPS based devices before and after applying 10 consecutive pulses. O1s XPS peak area ratios of (g) PDPP3T and (h) PDPP3T/NIPS based devices with and without consecutive pulse application. Operating mechanisms of (i) PDPP3T and (j) PDPP3T/NIPS in equilibrium, pulse on, and pulse off states.

equilibrium. However, because of the adsorption sites present on the channel polymer, some anions that accumulated in the channel layer did not immediately return to the electrolyte, resulting in a transient time. In addition to the existing adsorption sites, NIPS controls ionic transport, causing more anions to remain in the channel layer without diffusing back. In summary, PDPP3T/NIPS devices attracted more TFSI anions into the channel layer because of their high interfacial compatibility and suppressed the back-diffusion of adsorbed anions under the influence of crosslinking NIPS, resulting in a longer transient time and higher nonvolatile current characteristics.

Based on the improved synaptic properties of PDPP3T/NIPS devices, we investigated several biological functions of our devices as artificial synapses. Paired-pulse facilitation (PPF) and paired-pulse depression (PPD) refer to the facilitation or inhibition of neurotransmitter release through synapses by consecutive stimuli. The strength of facilitation for EPSC changes with the time interval (Δt) between the applied pair of pulses, which also works for the weakness of inhibitory

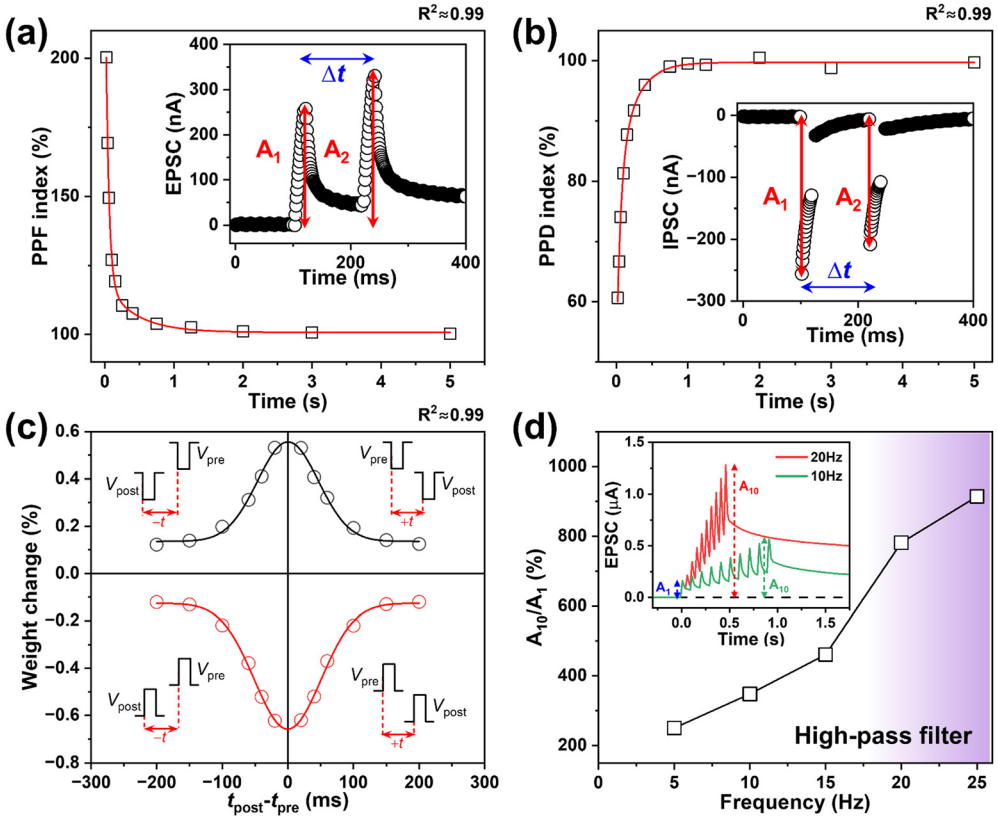


Fig. 4 (a) PPF index (A_2/A_1) according to the EPSC strength induced by two consecutive pulses (−2 V, 20 ms) applied with a time difference. Inset: EPSC strength measured at Δt = 100 ms and V_{DS} = −1 V. (b) PPD index (A_2/A_1) according to IPSC weakness induced by two consecutive pulses (2 V, 20 ms). Inset: IPSC weakness measured at Δt = 100 ms and V_{DS} = −1 V. (c) Symmetric Hebbian rule (black dot & line) and symmetric anti-Hebbian rule (red dot & line) behaviors recorded as a function of pre- and post-synaptic voltage applying time difference. The black dots were measured with two consecutive negative pulses (−2 V, 20 ms) and the red dots were measured with two positive pulses (2 V, 20 ms). (d) High-pass filter behavior with 10 consecutive pulses of 5 to 25 Hz frequency. Inset: EPSC under 10 consecutive pulses applying with 10 Hz (green line) and 20 Hz (red line) at V_{DS} = −1 V.

postsynaptic current (IPSC). Fig. 4a shows the EPSC response ratio of PDPP3T/NIPS upon applying a pair of negative pulses (−2 V, 20 ms) with different Δt values. Low Δt induces enhanced facilitation, resulting in higher EPSC upon applying the second pulse. Similarly, PPD shows suppressed current behavior as the Δt between a pair of positive pulses decreases (Fig. 4b). The PPF and PPD indices were determined as the ratios of the EPSC peak triggered by the first pulse (A_1) to the EPSC peak triggered by the second pulse (A_2).

$$\text{PPF \& PPD index} = A_2/A_1$$

The PPF/PPD index (black dot) gradually approaches 100% as Δt increases, that is, as the stimulus is applied and the time for the ions to return to their initial state is given. A double-exponential function was used to fit the PPF and PPD indices, indicated by the red curve. The double exponential function used for the fitting is

$$\text{PPF \& PPD} = C_1 \exp\left(-\frac{\Delta t}{\tau_1}\right) + C_2 \exp\left(-\frac{\Delta t}{\tau_2}\right) + C_0$$

where C_1 and C_2 are the initial accelerations of the slow and fast phases, respectively, and C_0 is a constant. τ_1 and τ_2 represent the relaxation times in each phase.[52] The values of τ_1 and τ_2 are 44 and 434 for PPF and 57 and 245 for PPD, which are similar to those of the human brain and reported synaptic devices, which are represented in tens and hundreds of ms.[53,54] To further confirm the low-power operation of our devices, we characterized the PPF of the PDPP3T and PDPP3T/NIPS devices in a V_{DS} value of −0.1 mV (Fig. S7, ESI†). When weak stimuli with V_{Pre} = −2 V and a width of 20 ms were applied, both PDPP3T and PDPP3T/NIPS devices exhibited PPF behaviors. To determine the energy consumption per spike, we performed the calculation using the equation below.

$$E = AIW$$

where A is the drain voltage, I is the drain current, and W is the electrical pulse width. The PDPP3T/NIPS device exhibited A = −0.1 mV, I = 294 nA, and W = 20 ms, which resulted in E = 0.588×10^{-12} J. The PDPP3T-based device measured in the same way recorded a drain current of I = 612 nA, resulting in an energy consumption per spike of 1.224×10^{-12} J, indicating that the device can mimic biological synapses with lower energy consumption after the introduction of NIPS.

Spike timing-dependent plasticity (STDP) is a property of learning and memory in which the strength of the connections between neurons is adjusted in response to the timing of

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stimulus application.[55] This is caused by changes in synaptic weights depending on the time interval between pre and postsynaptic spike firing. We applied a pair of programming or erasing pulses to the PDPP3T/NIPS devices as pre and postsynaptic spikes, respectively, to determine the synaptic weight changes as a function of the time interval. The synapse

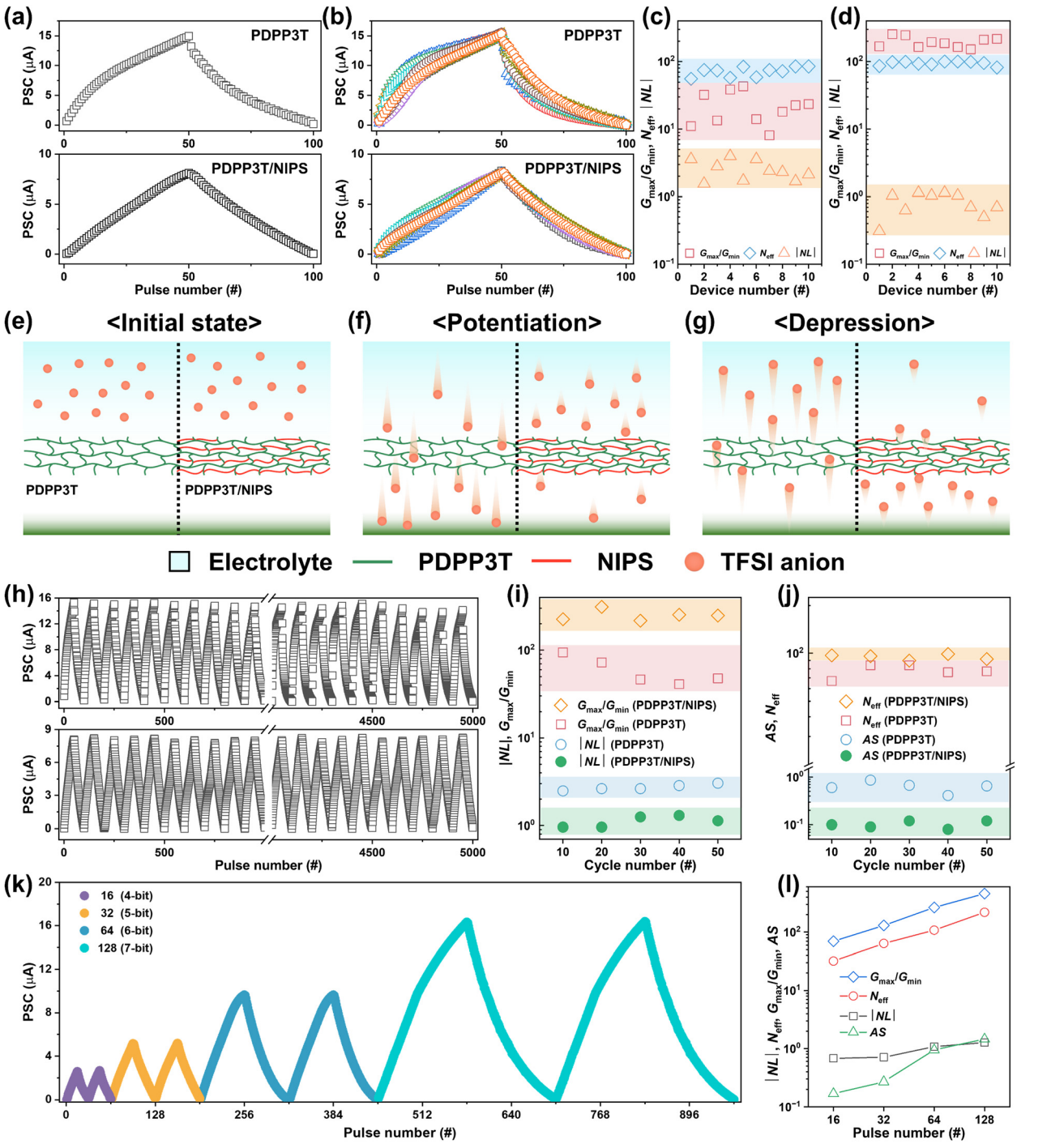


Fig. 5 (a) LTP/D characteristics of PDPP3T (upward) and PDPP3T/NIPS (downward) based devices by 50 programming/erasing pulses (V_{Pre} = −2 V/2 V, 80 ms width and interval) at V_{DS} = −0.7 V. (b) Device variations in a single cycle for 10 devices of PDPP3T and PDPP3T/NIPS based devices. G_{max}/G_{min}, N_{eff}, |NL| based on each (c) PDPP3T and (d) PDPP3T/NIPS based device. The ion transport properties of PDPP3T and PDPP3T/NIPS in the (e) initial, (f) potentiation, and (g) depression states. (h) Cycle variation of PDPP3T (upward) and PDPP3T/NIPS (downward) based devices. (i) G_{max}/G_{min} and |NL| in PDPP3T and PDPP3T/NIPS based devices as the cycle progresses. (j) N_{eff} and AS in PDPP3T and PDPP3T/NIPS based devices as the cycle progresses. (k) LTP/D according to multi (4,5,6,7)-bit operations of the PDPP3T/NIPS based device and (l) the recorded values of |NL|, N_{eff}, G_{max}/G_{min}, and AS.

weight change (ΔW) is defined as

$$\Delta W = \frac{I_{STDP} - I_0}{I_0} \times 100\%$$

I_0 is the current stimulated by the presynaptic spike and I_{STDP} is the current stimulated by the postsynaptic spike.[56] When the programming pulses (−2 V, 20 ms) were applied, ΔW increased, which satisfied the symmetric Hebbian rule of a gradual and symmetrical decrease by increasing the time interval of pre and postsynaptic spikes (black dot & line). On the other hand, ΔW decreased when the erasing pulses were applied, satisfying the anti-symmetric Hebbian rule that the degree of inhibition decreases with the increasing time interval of a pair of synaptic spikes (red dot & line). Our synaptic devices that satisfy the two Hebbian rules may exhibit LTP and LTD properties in terms of learning and memory.

As the last step in mimicking biological synapses, we implemented high-pass filtering to increase responsiveness to urgent signals by ignoring or reducing the intensity of signals below a certain frequency during the movement of neurotransmitters between neurons through synapses.[57,58] EPSC gains were calculated as the ratio of the 1st pulse (A_1) to the 10th pulse (A_{10}) when 10 consecutive pulses (−2 V, 20 ms) with a frequency of 5–25 Hz were applied to the PDPP3T/NIPS devices (Fig. 4d). The A_{10}/A_1 ratio increased rapidly from 461% to 782% in the 15–20 Hz interval. This confirmed that PDPP3T/NIPS devices can mimic high-pass filtering and have a filter for signals below 20 Hz when 20 ms pulses are applied.

Implementing artificial synapses for reliable neuromorphic computing requires a variety of characteristics in addition to mimicking biological synapses.[59] In the multistep weight update process, a low PSC facilitates power-efficient operation, and the linear behavior enables cross-tracking between the pulse number and PSC. In addition, a high dynamic range (G_{max}/G_{min}), a sufficient number of effective states (N_{eff}), and a low asymmetry degree (AS) are also important parameters. Fig. 5a shows the LTP/D characteristics of PDPP3T and PDPP3T/NIPS devices. We applied 50 programming pulses of −2 V and erasing pulses of +2 V with a width and interval of 80 ms at V_{DS} = −0.7 V, respectively. From the two curves, we can see that in addition to the PDPP3T/NIPS device having a lower PSC of 8.12 μA, it also has a more linear PSC modulation. To quantify linearity, we calculated nonlinearity (NL) using the following equations:[60]

$$G_{LTP} = B\left(1 - e^{\left(-\frac{P}{A}\right)}\right) + G_{min}$$

$$G_{LTD} = -B\left(1 - e^{\left(\frac{P-P_{max}}{A}\right)}\right) + G_{max}$$

$$B = \frac{G_{max} - G_{min}}{1 - e^{\left(-\frac{P_{max}}{A}\right)}}$$

G_{LTP} and G_{LTD} refer to the conductance of the potentiation and depression processes, respectively, and G_{min} and G_{max} refer to the minimum and maximum conductance, respectively. P_{max} is the number of pulses required to modulate G_{min} to G_{max}, and A is a parameter that controls the nonlinear behavior of the weight update. B is provided as a function of A to fit the conductance change curve within the ranges of G_{max}, G_{min}, and P_{max} to the given equation for NL extraction. The calculated |NL| was 1.88 in the LTP region and 2.19 in the LTD region for PDPP3T, while PDPP3T/NIPS improved to 0.54 in the LTP region and 0.83 in the LTD region (Fig. S9, ESI†). We examined G_{max}/G_{min}, N_{eff}, and |NL| across the 10 devices to determine whether the changes in the properties due to the introduction of NIPS were generalized (Fig. 5b–d). G_{max}/G_{min} was calculated as the maximum PSC value divided by the minimum PSC value, while the PDPP3T device averaged 28.5, which increased to 195.7 with the introduction of NIPS. N_{eff} defined the number of states in which ΔG (G_{n+1}–G_n) is greater than the noise during the conductance modulation process as half the value when the conductance is ideally modulated by applying the following equation.

$$G_{noise} = (G_{max} - G_{min}) \times \frac{1}{2P_{max}}$$

The noise due to the application of 50 programming/erasing pulses is assumed to be 1% of G_{max}–G_{min}. N_{eff} showed a significant increase in the number of distinguishable states out of 100, increasing from 72.3 to 94.3 with NIPS crosslinking. Finally, |NL| was compared using the lower values calculated in the LTP and LTD regions and was significantly reduced from 2.57 to 0.82 by NIPS crosslinking. These enhancements of the LTP/D properties were achieved by precise controlling of the ionic transport with NIPS. In the initial state, the PDPP3T/NIPS channel was not significantly different from the PDPP3T channel, except for the crosslinked structure owing to NIPS crosslinking (Fig. 5e). As shown in Fig. 5f, the TFSI anions accumulated in the channel upon negative pulse application. Because the PDPP3T channel does not control the movement of ions, TFSI anions quickly enter the loose polymer region, and nonlinear potentiation is recorded. In contrast, PDPP3T/NIPS channels allow the sequential accumulation of anions owing to their crosslinked structure; therefore, the potentiation process occurs sequentially depending on the pulses applied. The depression process was similar. The application of a positive pulse caused the anions that accumulated in the channel to return to the electrolyte. During this process, the PDPP3T channel experienced rapid dissipation, but the crosslinked NIPS inhibited the dissipation of ions, leading to sequential depression (Fig. 5g).

To ensure that our devices controlling the ionic transport can operate reliably over repeated operations, we evaluated their endurance by applying 5000 pulses (50 cycles) to a single device (Fig. 5h). For the first 10 cycles, the PDPP3T and PDPP3T/NIPS devices exhibited uniform behaviors. However, after 40 cycles, the PDPP3T device tended to exhibit a rapid increase/loss of synaptic weight during potentiation/depression, whereas the PDPP3T/NIPS device maintained similar performance. To further explore the cycle-to-cycle variation, we observed changes in the four parameters as the cycles progressed (Fig. 5i and j). PDPP3T/NIPS showed a 1.6 lower average |NL| and 191.1 higher dynamic ratio than PDPP3T. Furthermore, while PDPP3T showed an unsatisfactory N_{eff} value

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of up to 84 states, NIPS crosslinking improved it to at least 92 states. The AS, which represents the difference between |NL| values in the LTP and LTD regions, was also 0.54 lower on average for PDPP3T/NIPS devices, demonstrating superior performance and long-term durability. Utilizing the PDPP3T/NIPS devices with further improved performance, we conducted a multibit operation (Fig. 5k and l). We applied 16, 32, 64, and 128 pulses to simulate 4-, 5-, 6-, and 7-bit, respectively, and the PSC increased proportionally with the number of pulses to 2.61 μA, 5.16 μA, 9.63 μA, and 16.35 μA. While some synaptic properties tend to degrade as the number of applied pulses, an |NL| value of 1.27 and an N_{eff} value of 219 out of 256 states in 7-bit prove that our device still has good linear weight update characteristics.

We utilized convolutional neural network (CNN) and artificial neural network (ANN) simulations to determine the advantages of the better synaptic properties of PDPP3T/NIPS devices. The CNN was designed with a VGG-8 structure (six convolutional and pooling layers and two fully connected layers) based on a set of 10 groups of Canadian Institute for Advanced Research (CIFAR-10) test images (Fig. 6a). The input image, consisting of 32 × 32 pixels, was divided into three colors: red, green, and blue. In the convolution and pooling layers, interkernel convolution is used to form a feature map of the input image, and pooling is used to highlight and minimize the features and pass them to the next layer. After six processes, the fully connected layer compared the image pattern for each group with the input data and calculated the matching probability using the softmax function to record the classification accuracy. We performed recognition training with 50 000 randomly selected images and calculated the classification

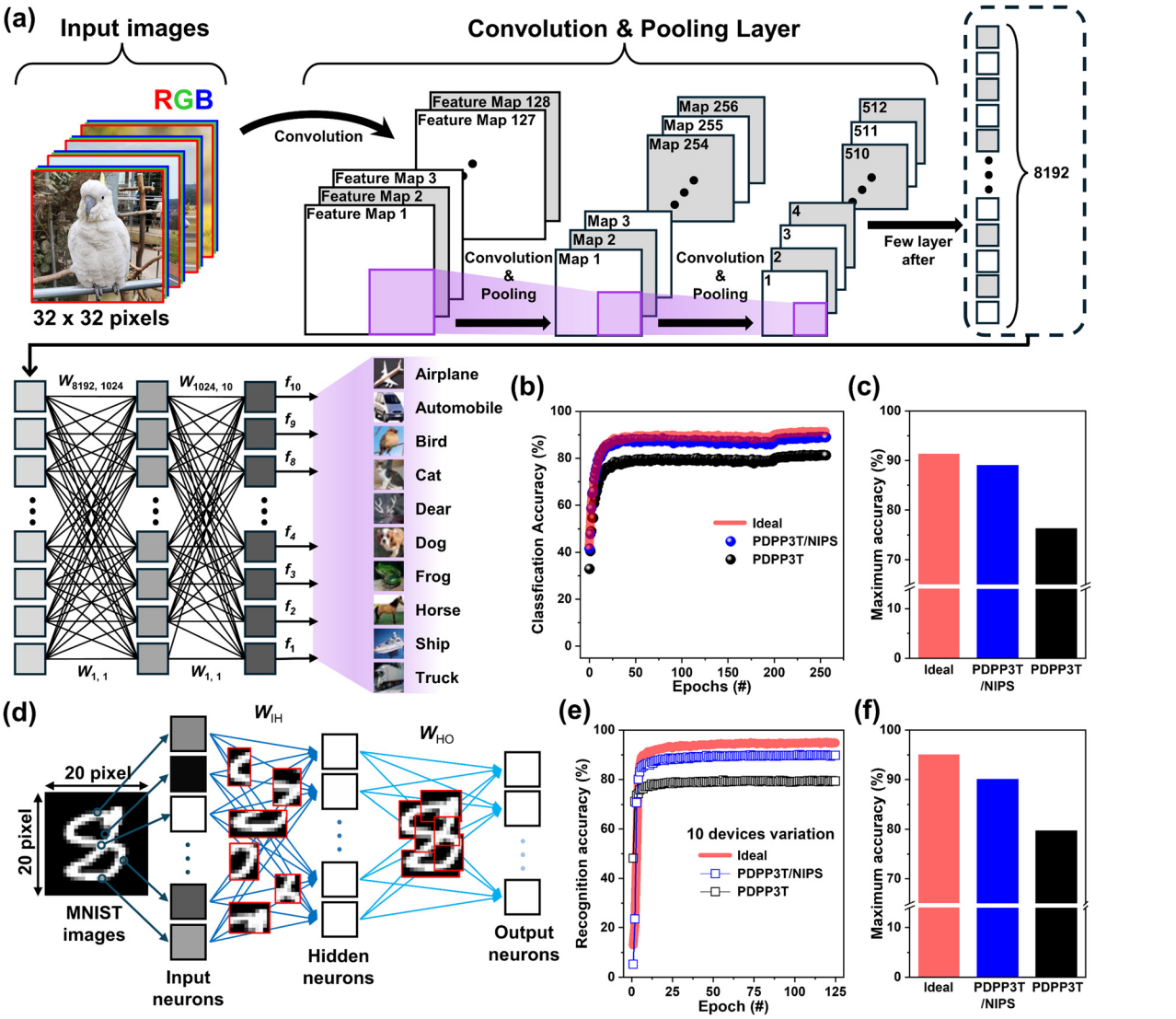


Fig. 6 (a) Artificial neural network structure with the VGG-8 structure for the CIFAR-10 based object classification. (b) Group classification accuracy over 250 epochs for 10 objects with PDPP3T, PDPP3T/NIPS and ideal devices. (c) Maximum classification accuracy for ideal, PDPP3T/NIPS, and PDPP3T devices. (d) Multilayer perceptron neural network structure containing 400 input neurons, 100 hidden neurons, and 10 output neurons for 20 × 20 input pixels. (e) Number recognition accuracy over 125 epochs in a device-to-device variation with 10 integrated devices, respectively. (f) Maximum recognition accuracy for ideal, PDPP3T/NIPS, and PDPP3T integrated systems.

accuracy of the CIFAR-10 image set with 10 000 images. The parameters used to evaluate the classification accuracy are the values of the PDPP3T and PDPP3T/NIPS devices shown in Fig. 5a, which is a hypothetical device that has most of the same parameters as the PDPP3T/NIPS devices, including the dynamic range, but the |NL| value is 0.01. As a result, the introduction of NIPS significantly increased the classification accuracy from 76% to 89%, which is very close to the value of 91% for an ideal device (Fig. 6b and c).

Low device-to-device variations are essential for neuromorphic chip engineering, which integrates multiple devices and drives them as a single system. To perform a parallel process, each device must record the same output for the same input. Deviations in parameters such as the dynamic range, |NL|, and maximum and minimum conductance between devices cause confusion during the process, leading to performance reduction. We tested the recognition accuracy of Modified National Institute of Standards and Technology (MNIST) handwritten digit images to simulate a multilayer perceptron-based ANN. The ANN is a two-layer perceptron consisting of 400 input neurons originating from 20 × 20 pixel images, with 100 hidden neurons and 10 output neurons representing digits 0 to 9 (Fig. 6d). The parameters used to evaluate the recognition accuracy were calculated from the |NL| and dynamic range of the average conductance per state of the ten PDPP3T and PDPP3T/NIPS devices shown in Fig. 5b, with the device variation applied by taking the average value of the ratio of the standard deviation to the conductance recorded per state (Fig. S13, ESI†). The ideal devices were based on the PDPP3T/NIPS device with the lowest |NL| and assumed a system integrating 10 completely identical devices (variation was zero) with most parameters, including the dynamic range, with an |NL| value of 0.01. After considering device variation, the ideal accuracy of 95% was reduced to 80% for PDPP3T, whereas PDPP3T/NIPS achieved 90% accuracy, a loss of only 5%. The properties of the PDPP3T/NIPS device were compared with recently reported synaptic devices (Table S2, ESI†). High linearity and excellent synaptic properties could be achieved by controlling the ion transport properties.

Conclusions

In conclusion, we developed a synaptic device with improved ion doping efficiency and ion transport control properties *via* click chemistry-designed photocrosslinking. Oligo(ethylene glycol) introduced *via* crosslinking enhanced the electrolyte/channel interface compatibility without causing crystallinity degradation, which improved the ion doping efficiency and effectively inhibited the back-diffusion of accumulated ions. The improved synaptic properties enabled the realization of biological synaptic features such as STDP and high-pass filtering. The linear weight update, high dynamic range, and excellent endurance exhibited by the LTP/D process demonstrate that crosslinked NIPS can effectively control ion transport. The high accuracies of 89% and 90% recorded in CNN and ANN simulations also support the advantages of NIPS crosslinking. This study, which improved the ion doping efficiency and ion transport control properties of OECT-based synaptic devices *via* photoinduced crosslinking, opens up the possibility of constructing a new generation of artificial synapses through an innovative approach to polymer structure modulation for synaptic property enhancement.

Experimental section

Please refer to the ESI† for details on the experimental procedure, device fabrication and characterization.

Author contributions

H. J., G. Y. B., and E. L. introduced novel artificial synapses based on PDPP3T and NIPS. H. J. and J. S. designed and fabricated the devices. G. Y. B. and S. H. K. performed material selection and synthesis method optimization. J. S. performed neural network simulations. H. J. and J. S. wrote this manuscript based on discussions with all authors. E. L. supervised the project, including the experimental and theoretical background of this study.

Data availability

The data supporting this article have been included as part of the ESI.†

Conflicts of interest

There are no conflicts to declare.

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